The Network of Network Science

An Analysis of Co-Authorship

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Abstract

The field of network science is clearly defined by its interdisciplinary nature, drawing knowledge from physics, computer science and social science. Co-authorship networks are a reliable measurement of such scientific collaboration by constructing a social network of researchers. We investigate a co-authorship network of the field of network science to answer the question how knowledge and researchers are organized within the field. To this end we employ several metrics such as centrality measures, degree distribution, community detection and robustness analysis. The results indicate a network with scale-free and small world characteristics. Researchers seem to organize into small communities which are connected over a few important connecting researchers.

Introduction

Scientific collaboration is the backbone of academia and has significantly increased over the last 25 years (Ullah et al. 2022), because collaborative research provides findings beyond what one researcher is capable of and is the result of integral parts of modern science such as interdisciplinary research, mentoring relationships and globalization (Dusdal and Powell 2021). Co-authorship networks are one of the most reliable measurements to investigate scientific collaboration (Ullah et al. 2022; Kumar 2015). These networks represent a social structure of researcher, which gives us a better understanding of the self-organization rules of science and lets us investigate research teams and key research partners (Molontay and Nagy 2021; Ullah et al. 2022). The field of network science in particular is interesting to analyze in such a way, because of its highly interdisciplinary origin and widespread application (Molontay and Nagy 2021). This exploratory project tries to answer the following research question: How is the academic field of network science organized as measured by a co-authorship network? To this end we will analyze the co-authorship network provided by Rossi and Ahmed (2015) in regards to its basic structure, centrality measures, robustness and communities.

Results

Degree and Clustering: The network has 379 nodes and 914 edges and is undirected. None of the nodes have attributes, meaning we only concentrate on structural features formed by the edges. The network density is 0.013, meaning only 1,3 percent of all possible connections are actually realized, qualifying the network as a sparse network. The average shortest path length is 6.042. This indicates a small-world property, which we tested statistically by simulating a random graph as null-model. We computed the small-world coefficient as $\sigma = \frac{C/C_r}{L/L_r}$, where C and L are the average clustering coefficient and average path length of the original graph, and C_r and L_r of the random one (Humphries and Gurney 2008). With $\sigma = 10.273 > 1$, we can confirm the small world property. The global clustering coefficient (0.4306) being smaller than local clustering coefficient (0.7412), suggests the network has a tendency to form individual communities which are connected over some individual notes. Figure 1 the distribution resembles a scale-free network with few highly connected nodes

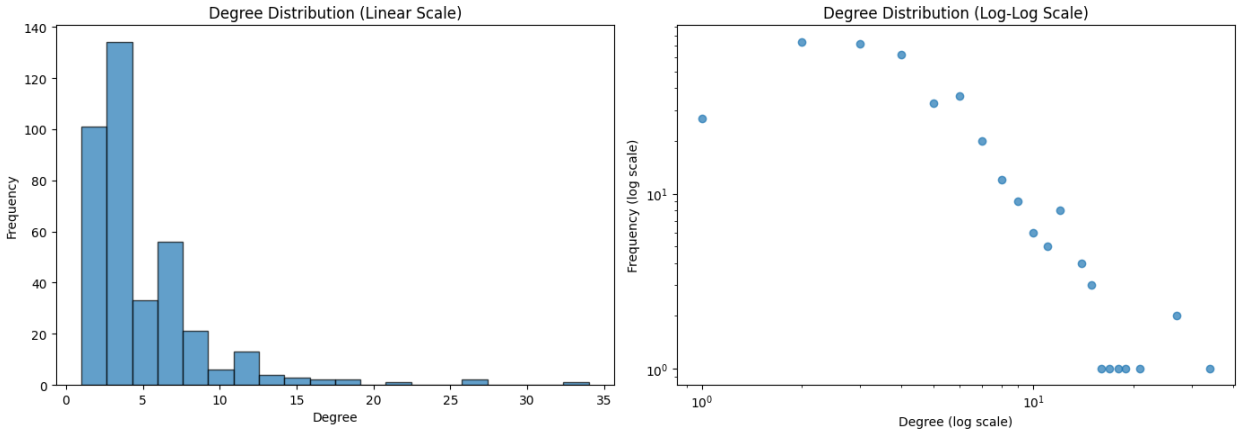

Figure 1: *Left. Degree Distribution in Linear Scale.Right. Degree Distribution in Log-Log Scale.*

many low connected nodes. This notion is further supported by a degree assortativity of -0.0817, where high-degree nodes tend to connect more to low-degree nodes.

Centrality measures: We analyzed and compared various centrality measures such as degree, betweenness, closeness, eigenvector centrality and clustering coefficient. Figure 2a shows tha degree is positively associated with betweeness, closeness and eigenvector centrality. This means that a higher number of edges is generally associated with a more central position within the network (closeness), where the node connects many different nodes (betweeness) and is also well connected to other well connected nodes (eigenvector). The clustering coefficient shows a negative correlation with each other centrality measure, because the more connections a node has, the more connections have to be closed to get a high clustering coefficient. The association with eigenvector centrality, while negative is rather weak, suggesting that nodes which are only connected to other well connected nodes can be part of highly clustered communities. These results support the notion that the network is organized in small communities, which are connected by a few high betweenness nodes.

Robustness: We use the aforementioned centrality measures further in a robustness analysis. We simulated random and targeted attacks on the network. Targeted attacks were done using the highest nodes by degree, betweenness, closeness and eigenvector centrality. For each simulation we removed stepwise up to 10 percent of the nodes within the network. Figure 2b shows the the fraction of nodes in the largest connected component after removing a certain number of nodes. The network is more resilient to random attacks than targeted attacks, indicated by the large fraction of nodes still remaining in the giant component after removing 10 percent of nodes randomly. The Molloy-Reed criterion at 8.021 and the critical threshold at 0.8576 confirms this robustness against random attacks. Therefore we would have to remove around 85,76 percent of the nodes to make the giant component break down completely. When targeting nodes by betweenness, degree and closeness, the giant component breaks down after the first node is removed. Targeted attacks by betweenness and closeness lead to an great initial reduction of the fraction of nodes in the giant component, followed by a steady decline. Degree based attacks break apart the giant component slower but there is a threshold at which the fraction of nodes in the giant component decrease greatly. Betweenness based attacks fragment the network more than closeness and degree based attacks, because it targets nodes, which are more likely to be on the shortest path between two nodes, serving as impor-

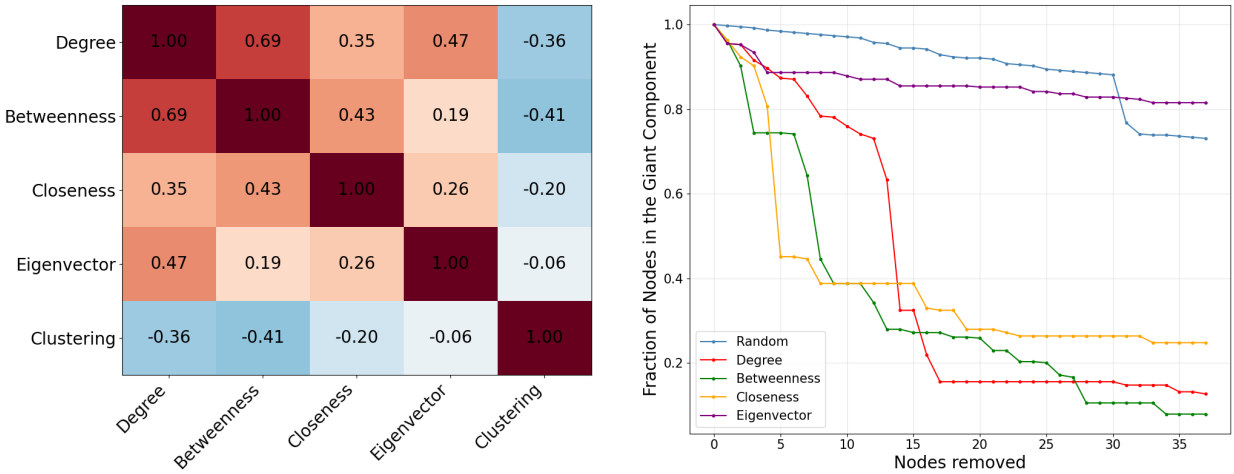

Figure 2: *Left. Pearson Correlation Matrix of Centrality Measures.* The color scale denotes the orientation of correlation with shades of blue for negative and shades of red for positive correlations. *Right. Robustness Analysis.* The different colors and lines denote different centrality measures by which we removed up 10 percent of nodes stepwise when simulating targeted attacks.

tant connectors. Targeted attacks based on eigenvector centrality behave similarly to the random attacks, because nodes with high eigenvector centrality are not necessarily the most well connected within a network. They are connected to other well connected nodes, so removing them on their own does not impact the network the same way other centrality measures do. The robustness against random failures but vulnerability against targeted attacks is an additional characteristic of a scale-free network we have already established.

Community detection: We found 19 distinct communities and on average 19.9 researchers belong to one community as determined by the Leiden algorithm. Furthermore, we calculated a modularity of 0.8486, which supports our observation and earlier claim about communities being well connected within but not across communities. Figure 3 shows an interconnected core of the network on the right hand side where communities are not clearly separated. Spreading outwards, communities show a tree-like structure where one node is connected to many individual nodes. In a similar manner, some communities seem to be connected to the main network solely over other communities, as observable at the upper left corner, where the black and pink community seem to originate from the coral community or the lower left corner, where the yellow community connects the rose and brown community to the network. To summarize, the result indicate a scale-free network according to its degree distribution and robustness analysis. It shows characteristics of a small world-property determined by path length and clustering coefficients. It is organized into small communities which are connected over few important nodes to one another to form the giant component as supported by the centrality measures and visualization.

Discussion

These results allow us to characterize how the field of network science is organized. The scale-free structure suggests that there are a small number of highly connected researchers, which serve as bridges between highly clustered communities, indicated by the collapse of the giant component under targeted attacks. Some of these highly connected researchers are the main connection for sub-communities to the main

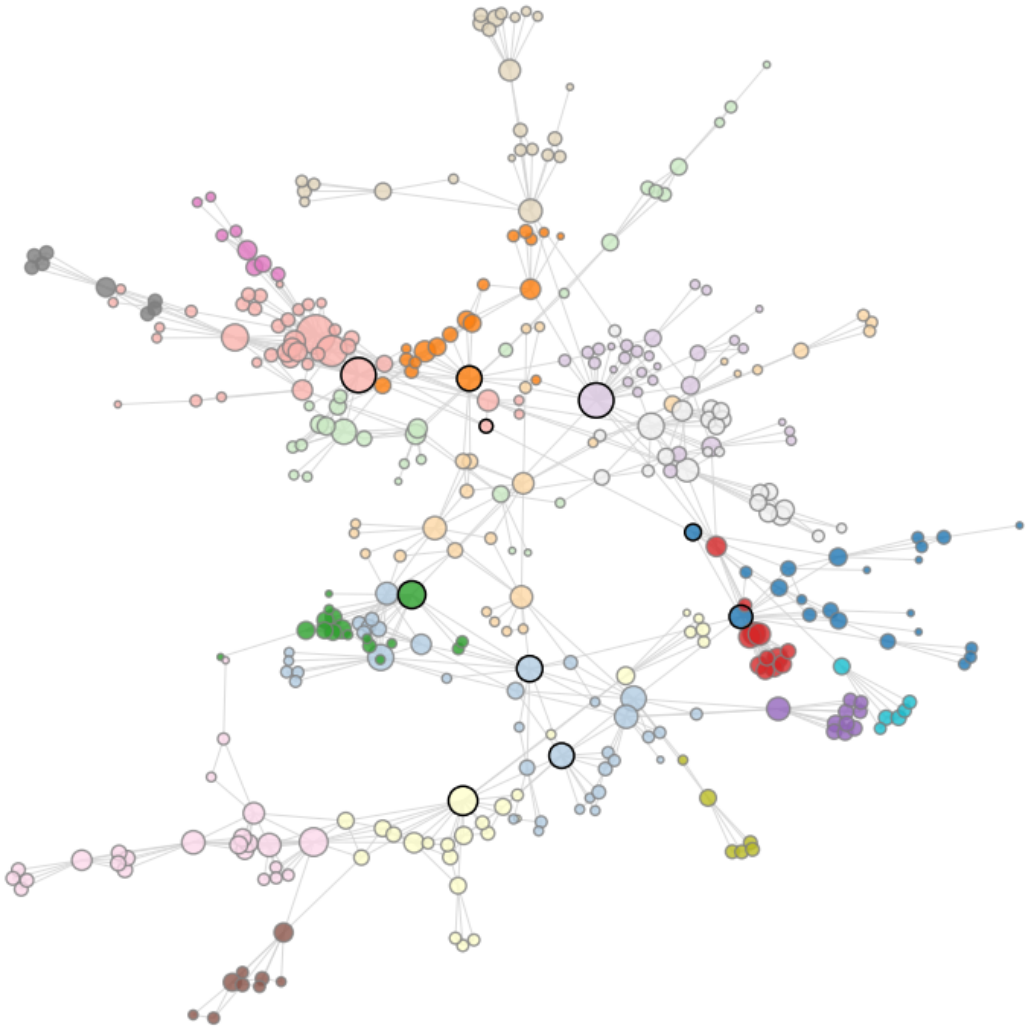

Figure 3: *Network Visualization.* Colors denote communities as determined by the Leiden algorithm, which found 19 distinct communities. Node size is proportional to degree. Black outline represents the ten highest betweenness nodes within the network.

network, meaning they are either a different discipline doing interdisciplinary research or are originating from a main branch following different approaches to the same problem. These sub-communities could also be departments, countries or research institutes. The tree-like structure we observed might be indication of student and supervisor relationships. We showed with the random null model that the network has a small world property, which means that two researchers are able to collaborate with each other through six co-authorship links on average. However the network diameter of 17 suggests, that this varies across the network, meaning the small world property is likely maintained by the highly connected researchers. Our results are consistent with the current literature on co-authorship networks. Barabási et al. (2002) characterized co-authorship as scale free networks and Newman (2001) as having a small world. According to Kumar (2015) these two properties are related to one another, because being small world requires popular nodes within a network, which is a sign of a scale-free network. A small property additionally requires a high clustering coefficient (Newman 2001). The lower global clustering coefficient is much smaller compared to other analyses Kumar (2015) and Molontay and Nagy (2021). This might be because the data used in those papers is much richer than the network we investigated here. For example, the network investigated by Molontay and Nagy (2021), contained all authors which cited at least one of three predefined central papers within the network science community resulting in 56.646 scientists included in the network. The data we are working with could be subset or a network with stronger restrictions which might explain the inherent difference in results. A main limitation of this analysis is the absence of node attributes. Without information about the researchers, departments and disciplines a deep interpretation of the communities is more difficult as already mentioned by Kumar (2015) and Molontay and Nagy (2021). Future work should address these shortcomings by including such data.

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